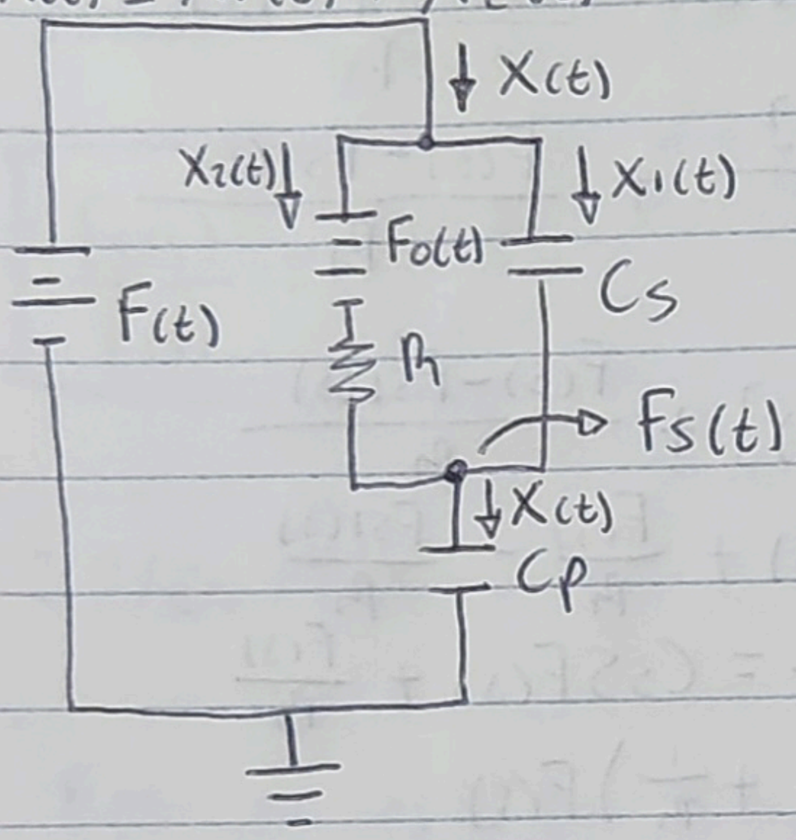
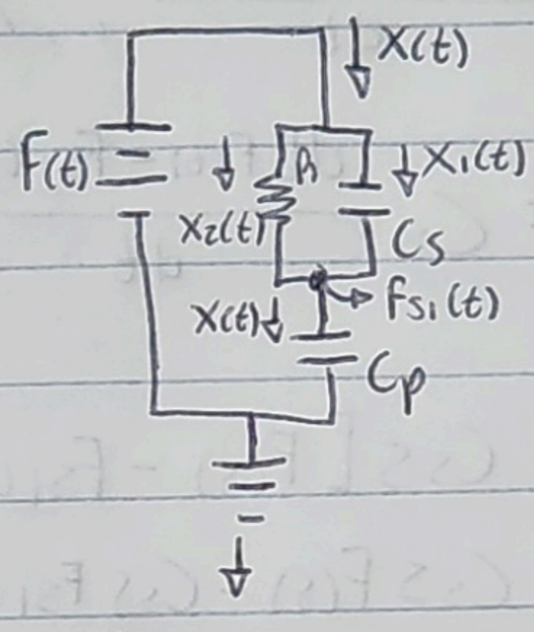
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Práctica 3. Modelado y control del sistema musculoesquelético.

$X(t) = X_1(t) + X_2(t)$



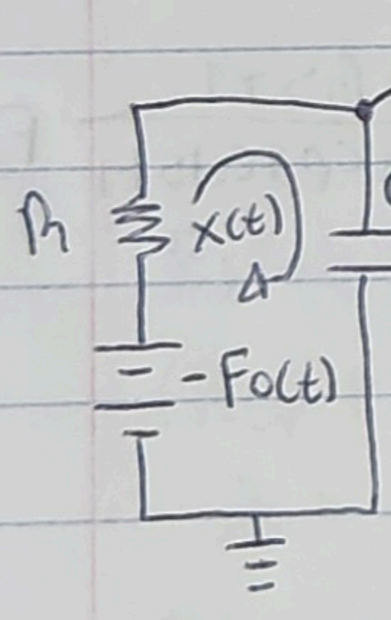
1er Cto.



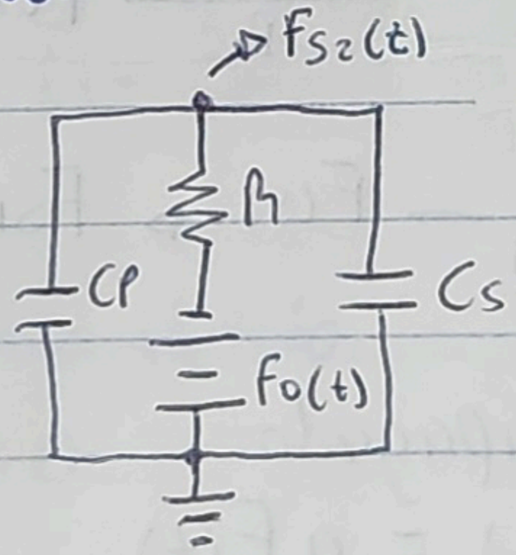
2do Cto.

Superposición

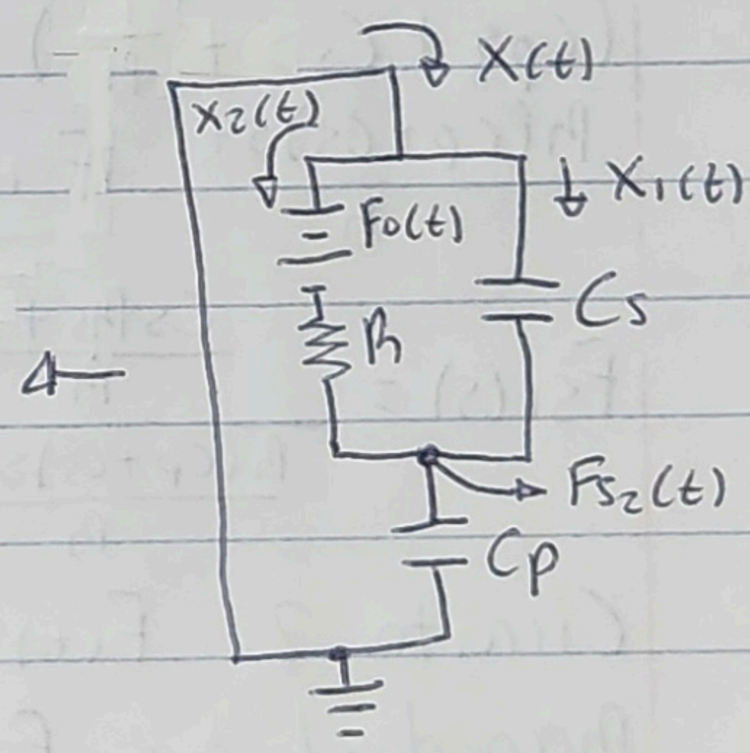
$F_S(t) = F_{S1}(t) + F_{S2}(t)$



=



+



Análisis matemático

Debido a que el sistema tiene dos fuentes de voltaje, se debe aplicar el principio de superposición.

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Circuito 1. $F_0(t)$ apagado.

$$x(t) = x_1(t) + x_2(t) \; ; \quad x(t) = C_p \frac{d F_{s_1}(t)}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} \; , \quad x_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p \frac{d F_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p s F_{s1}(s) = C_s s [F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) = C_s s F_{s1}(s) + \frac{F_{s1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} \boxed{F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)}$$

$$F_{s1}(s) = \frac{\frac{C_s R s + 1}{R}}{\frac{R(C_p + C_s)s + 1}{R}} F(s) \rightarrow F_{s1}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

Circuito 2. $F(t)$ apagado.

Recordando que $F_0(t) = a F(t)$

$$-a F(t) = R x(t) + \frac{1}{C_p + C_s} \int x(t) dt$$

$$f_{s2}(t) = \frac{1}{C_p + C_s} \int x(t) dt$$

$$F(s) = -a f(s) = R X(s) + \frac{X(s)}{(C_p + C_s)s} = \frac{-1}{\alpha} \left(R + \frac{1}{(C_p + C_s)s}\right) X(s) = -\frac{1 \cdot R(C_p + C_s)s + 1}{\alpha (C_p + C_s)s \cdot X(s)}$$

$$F_{s_2}(s) = \frac{X(s)}{(C_p + C_s)s}$$

$$\frac{F_{s2}(s)}{F(t)} = \frac{\frac{X(s)}{(C_p + C_s)s}}{-\frac{1}{\alpha} \cdot \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s} X(s)} = -\frac{\alpha}{R(C_p + C_s)s + 1}$$

$$\boxed{F_{s2}(s) = \frac{\alpha F(t)}{R(C_p + C_s)s + 1}}$$

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Función de transferencia.

$$F_S(s) = F_{S1}(s) + F_{S2}(s) = \frac{C_s R_s + 1}{R(C_p + C_s)s + 1} F(s) - \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$F_S(s) = \frac{C_s R_s + 1 - \alpha}{R(C_p + C_s)s + 1} F(s)$$

$$\boxed{\frac{F_S(s)}{F(s)} = \frac{C_s R_s + 1 - \alpha}{R(C_p + C_s)s + 1}}$$

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Error en estado estacionario.

$$e(s) = \lim_{s \to 0} s F(s) \left(1 - \frac{F_S(s)}{F(s)}\right) = \lim_{s \to 0} s \frac{1}{s} \left(1 - \frac{RC_s s - 1 - \alpha}{(RC_p + RC_s)s + 1}\right) = \alpha$$

Estabilidad en lazo abierto.

$$(RC_p + RC_s)s + 1 = 0$$

$$\lambda = -\frac{1}{RC_p + RC_s}$$

- El sistema es estable